N Mithun

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.32
Sri Chaitanya Techno School	Class XII (CBSE)	2024	91.60%
Sri Chaitanya Techno School	Class X (CBSE)	2022	95.60%

SKILLS

Programming: Python, R, C, C++ | **Data & ML:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

Deep Learning: PyTorch, TensorFlow, Keras | **CV:** OpenCV | **NLP:** spaCy, Hugging Face Transformers

Deployment & Tools: FastAPI, Streamlit, Git

RESEARCH EXPERIENCE

Undergraduate Researcher -- Computational Biomedical Engineering | *IIT Hyderabad*

- Conducting computational research in biomedical engineering involving **mathematical modeling, scientific programming, and computational analysis**.

RELEVANT COURSES

BSc Degree in Data Science and Programming - IIT Madras

Intro to Programming, Probability and Random Variables, Foundation of Machine Learning, DSA, Algorithms

PROJECTS

Brain Tumor & Brain Imaging Segmentation

- **Built a deep learning segmentation pipeline** using a U-Net-style encoder-decoder architecture in PyTorch to automatically segment brain tumors from multi-modal MRI scans.
- **Improved diagnostic consistency** by enabling faster, more consistent radiological assessment and reducing manual annotation effort for clinicians.
- **Strengthened model robustness** by applying preprocessing techniques such as skull-stripping, intensity normalization, and data augmentation to improve performance across varied scans.
- **Validated segmentation accuracy** by evaluating output quality using Dice coefficient and IoU metrics benchmarked against ground-truth tumor masks.

Surgical Tool & Joint Tracking (Computer Vision)

- **Built a real-time computer vision system** using OpenCV and a YOLO-based object detection pipeline to track surgical tools and joint movements from video.
- **Enabled motion analysis workflows** by supporting surgical workflow review through continuous, real-time tracking of tool and joint positions.
- **Maintained tracking consistency** by combining frame-by-frame detection with pose-estimation and tracking techniques to preserve stable identity of tools and joints across video sequences.
- **Evaluated system reliability** by analyzing tracking accuracy and stability under varied lighting conditions and occlusion scenarios.

Clinical NLP + FHIR Assistant

- **Built an NLP assistant** using spaCy and Hugging Face Transformer models to process unstructured clinical notes.
- **Automated structured data conversion** by transforming free-text clinical notes into structured, FHIR-compliant records, streamlining data entry for healthcare systems.

- **Extracted clinical entities** by performing named-entity recognition to identify key terms such as conditions, medications, and observations.
- **Ensured healthcare interoperability** by mapping extracted fields onto standard FHIR resources (e.g. Condition, MedicationStatement, Observation) for compatibility with existing healthcare data systems.

Real-Time Fraud Detection System (MLOps Pipeline)

- **Built a fraud detection model** using XGBoost/LightGBM trained on the ULB Kaggle credit card fraud dataset (~285K transactions, ~0.17% fraud rate).
- **Handled severe class imbalance** by applying SMOTE and class-weighting techniques, and optimized the model for AUPRC rather than plain accuracy.
- **Deployed the model as a live service** by serving real-time predictions through a FastAPI backend.
- **Enabled real-time monitoring** by building a Streamlit dashboard to track model performance and fraud detection metrics.

WORK EXPERIENCE

Data Analyst Intern -- L&T Shipbuilding, Chennai

- Performed **data cleaning and validation** on engineering measurement datasets, identifying inconsistencies and improving data accuracy.
- **Organized, transformed, and standardized raw measurement and cable data** to create structured datasets suitable for analysis and reporting.
- Analyzed **cable dimensions, specifications, and cost-handling constraints** to calculate final cable laying distances for engineering applications.